

Course

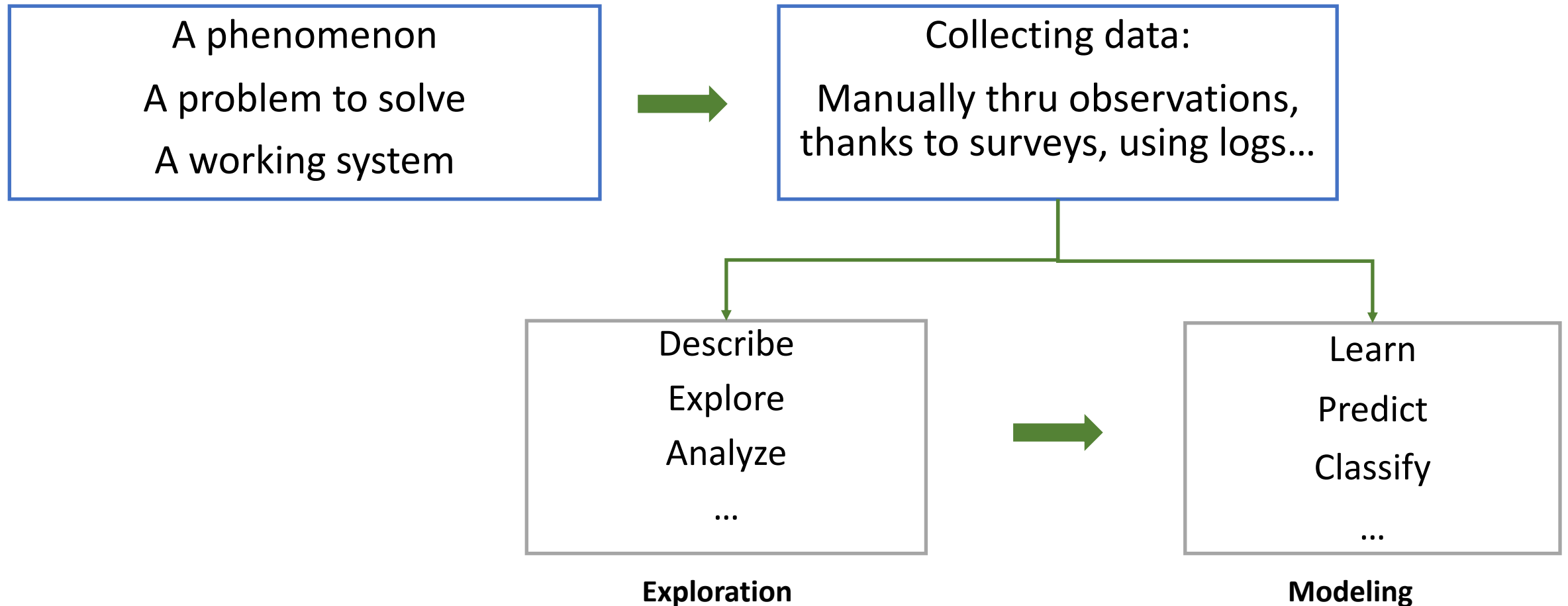
« *Deep Learning* »

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Introduction



A brief introduction to data exploration and analysis

- 1. High dimensional data*
- 2. Principal component analysis*
- 3. Unsupervised classification (clustering)*

Introduction

- Descriptive analysis:
 - Exploration
 - Data description
 - Focus on a sample
- We do not assume any prior model.
- As soon as we would like to generalize the analysis to all the data:
→ Modeling

Generalization



Modeling

Introduction

- **Example:**
Data about cars.

Description:

A automobile rental company would like to know the condition of its fleet.

- Diesel cars have a higher mileage than gasoline cars.

Modeling:

Used car sales website.

- Estimate the price of a car from its characteristics.

Introduction

- **Example:**
Covid-19: Data of a set of people is available

Description:

Analyze, describe, explore...

People with chronic diseases (such as diabetes or HTA) develop more acute symptoms.

Introduction

- **Example:**
Covid-19: Data of a set of people is available

Modeling :

Prediction, classification... towards generalization.

If you consider, age, socio-professional category,
family situation...

What is the probability of being infected with
Covid-19?

Introduction

- Descriptive analysis concerns a set of individuals: A population.
- Individuals are not necessarily people:
 - objects;
 - animals;
 - images;
 - videos;
 - sensor surveys;
 - ...
- And we do not study the population as a whole.
 - You can « observe » everything.
 - The study most often focuses on a sample.

Introduction

- Studying an individual/object => analyze its features/properties.
 - One feature → a variable
- The sample to be studied is represented as a table:
 - Rows = individuals
 - Columns = variables.

Sujet	Math	Sciences	Français	Latin	Musique
Jean	6	6	5	5,5	8
Aline	8	8	8	8	9
Annie	6	7	11	9,5	11
Monique	14,5	14,5	15,5	15	8
Didier	14	14	12	12	10
André	11	10	5,5	7	13
Pierre	5,5	7	14	11,5	10
Brigitte	13	12,5	8,5	9,5	12
Evelyne	9	9,5	12,5	12	18

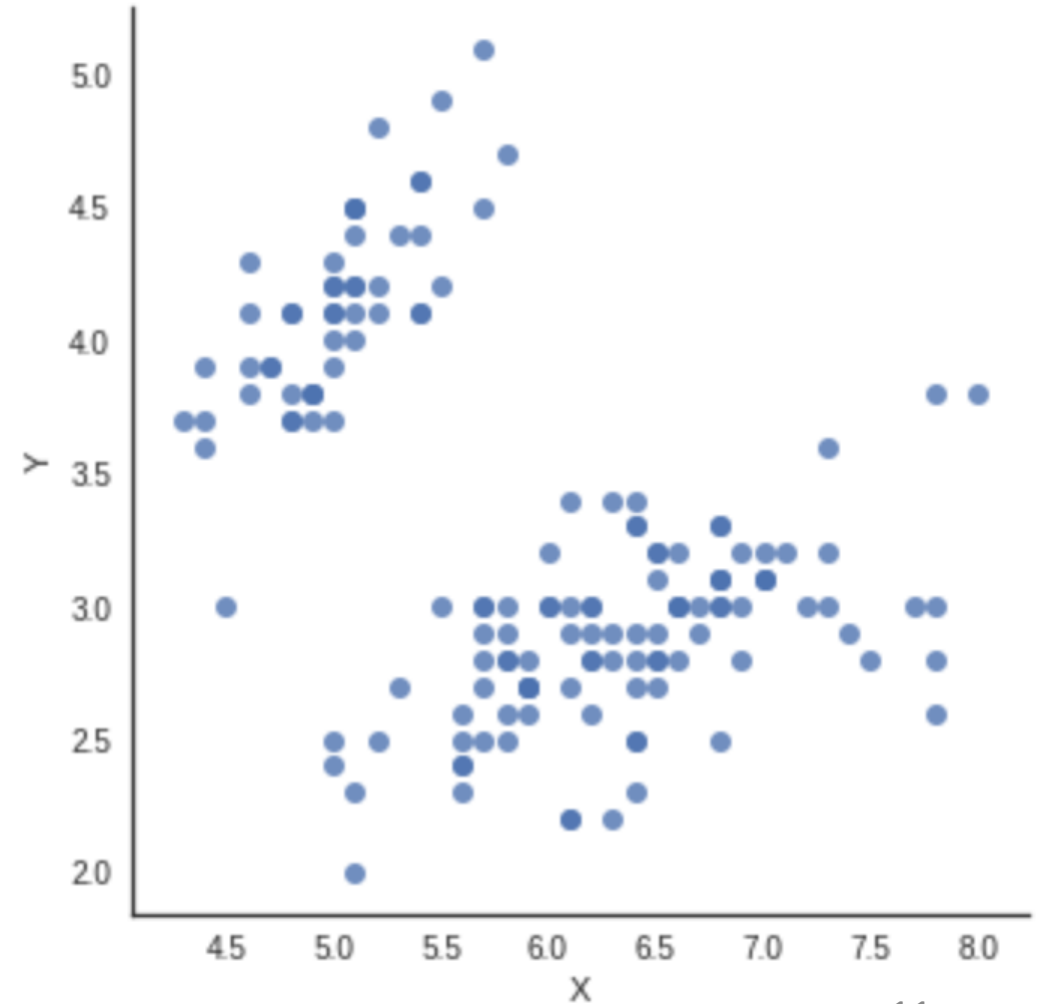
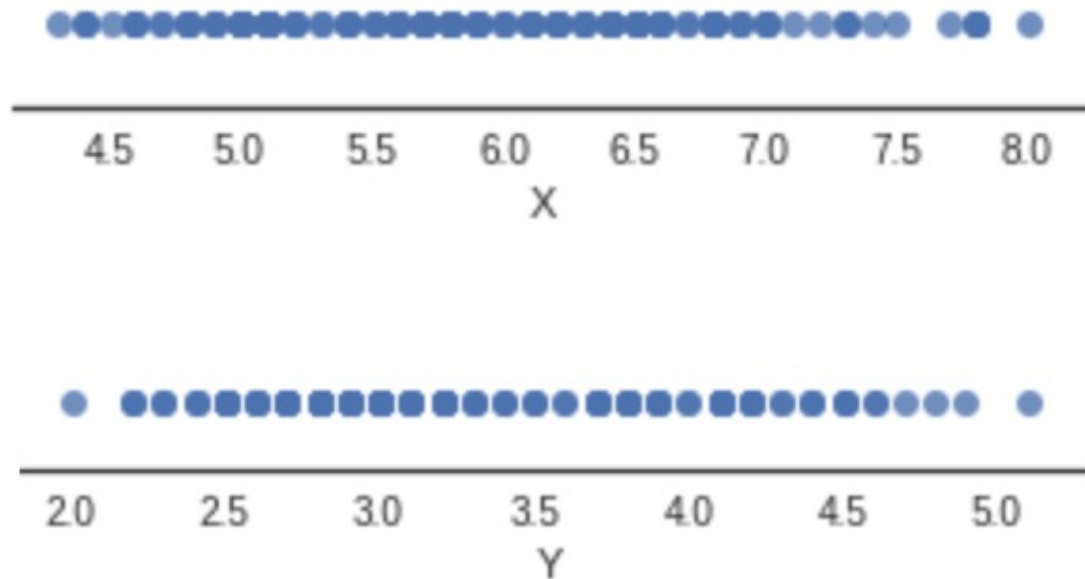
1. High-dimensional data

- One-dimensional analysis
=> Study variables one by one, and separately.
- With two variables
=> we can see, for example, the correlation.
- More variables analyzed jointly: multidimensional analysis.
=> Allows you to see more things!

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1. High-dimensional data

- Individuals: $I(X,Y)$



1. High-dimensional data

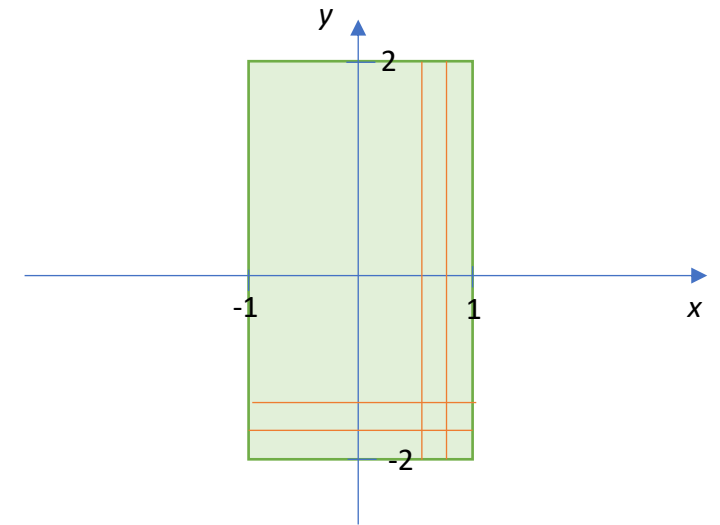
- Dimension > 3 : Brain unable to perceive and represent data.
- High-dimensional data has properties that are beyond our intuition.

→ *Dimensionality Curse.*

1. High-dimensional data

Dimensionality curse

- *This refers to the difficulties that arise in high-dimensions when it comes to:*
 - *Analyze,*
 - *Explore,*
 - *Make use,*
 - *Manage,*
 - *Process,*
 - *...*
- Introduced for the first time by Richard Bellman [1961]:
It refers to the difficulty to perform optimization in high-dimensional spaces using exhaustive methods.



1. High-dimensional data

Dimensionality curse

- Also used in statistics:

It refers to the relationship between the size of the data sample and the reliability of the estimate [Donoho 2000].

- It has been shown that the number of samples must increase *exponentially* with the dimension in order to maintain the same level of reliability of the estimate.

1. High-dimensional data

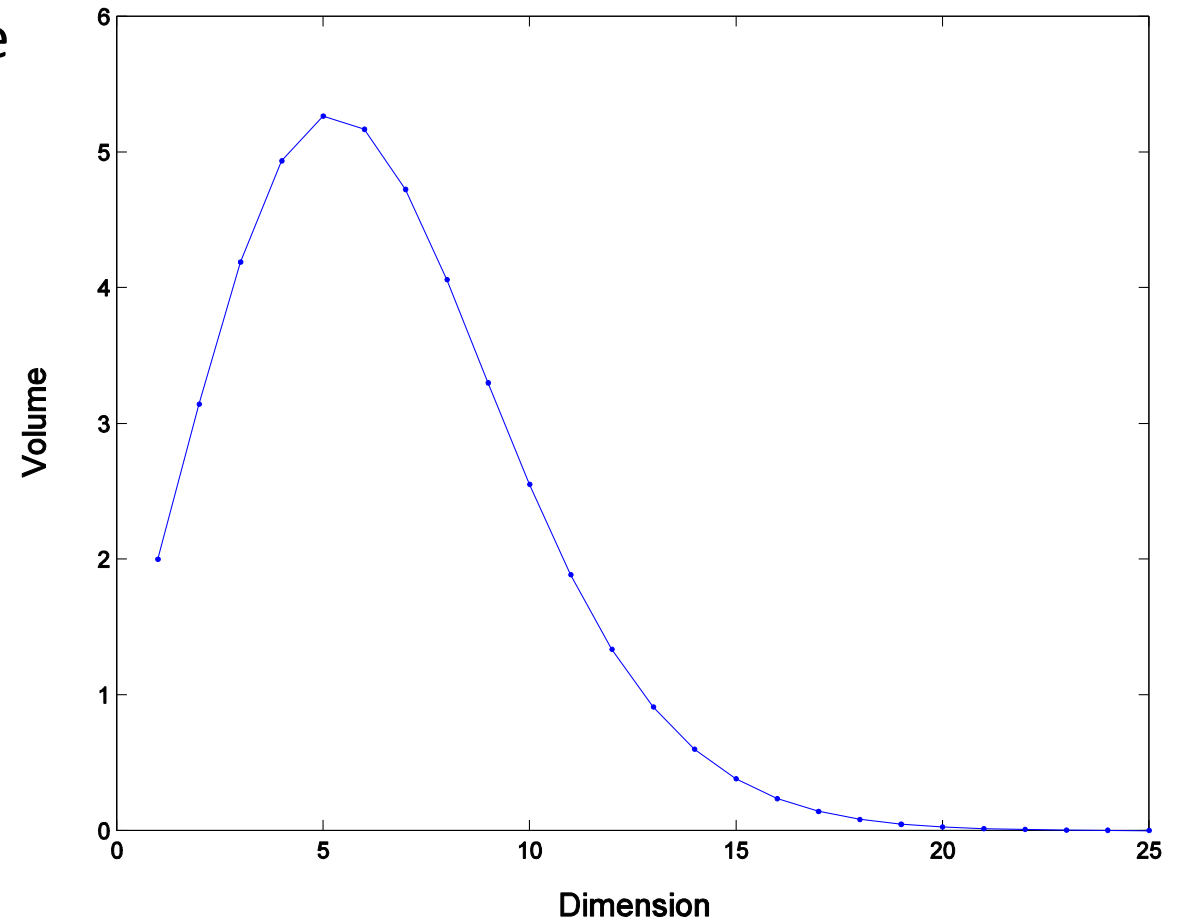
Dimensionality curse

- When the dimension increases:
 - The distance to the nearest neighbor increases
 - The distance to the furthest neighbor increases
 - $d(\text{nearest neighbor})$ increases faster than $d(\text{furthest neighbor})$
- Under certain conditions (including data uniformly distributed on all dimensions and independent):
 - When $d \rightarrow \infty$:
Distance to the nearest neighbor = distance to the furthest neighbor!
[Beyer 1999]

1. High-dimensional data

Dimensionality curse

- The volume of a hyper-sphere



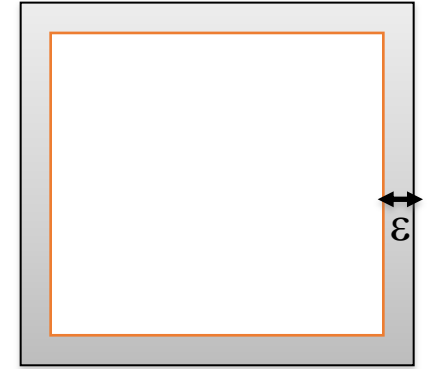
1. High-dimensional data

Dimensionality curse

Exercise

Let's consider a set of vectors uniformly distributed in a "hyper-square" (a 2D square, a 3D cube...).

Let's consider the space inside the hyper-square at a distance ε from the edge: grey space (in the figure).



- If we draw a random point inside the hyper-square, what is the probability that the point falls in the gray space, if we consider that the space is d -dimensional?
- Study the variation of this probability according to the dimension: 2, 3, 4,..., 100. Draw the corresponding curve.
- What do you notice?

1. High-dimensional data

A summary

- Number of variables (dimension) +++
- Number of individuals +++
 - => The analysis becomes difficult.
 - => Less or no intuition
 - => Costly processing

Two possible approaches:

1. Reduce the number of variable: Create new variables from the original ones while losing as little information as possible. **PCA**
2. Partition the data set (individuals) by grouping similar individuals **Clustering**